



■ CS177 Bioinformatics: Software Development and Applications

Lecture 0: Introduction to Course

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Outline



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Instructors



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- Lecturer

- Name: Jie Zheng (郑杰)
- Email: zhengjie@shanghaitech.edu.cn
- Phone: 21-2068-4861
- Office room: SIST Building 3, Room 432
- Office hour: Friday 4 – 5 pm, or by appointment



- Teaching assistants (TAs)

- Ms. Siyu Tao (陶思宇), Email: taosy2022@shanghaitech.edu.cn
- Mr. Ruoyin Zhang (张若隐), Email: zhangry2@shanghaitech.edu.cn

- Roles of TAs:

- Check attendance
- Answer technical questions
- Collect and correct homework assignments, and give feedbacks
- Help lecturer supervise and evaluate course projects



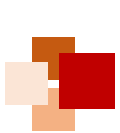
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Learning objectives



- **Bioinformatics** is basically the application of information science and technology to molecular biology (more introduction in Lecture 1 and Lecture 2)
 - Mainly about the processing, curation and analysis of biological data
 - The primary goal of bioinformatics is to increase our understanding of biological processes using computational methods and software tools
- After taking this course, you should be able to:
 - Formulate challenges and problems in bioinformatics as data science tasks
 - Describe and use algorithms and software tools well-known in the field to analyze biological data
 - Build machine learning models for biological data mining and prediction
- Note: This course is only a **basic introduction**. To become a bioinformatician or data scientist, you need a lot of practice.





Learning plan

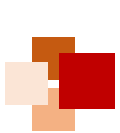


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Schedule	Teaching Content	Teaching Form
Week 1	1. Introduction to the course (objectives, policies, etc.) 2. Basics of molecular biology 3. Overview of bioinformatics	Lecture, discussion
Week 2	1. Basics of algorithms, probability and statistics 2. Basics of data science and machine learning 3. Markov models and Hidden Markov models (HMMs)	Lecture, discussion
Week 3	1. Overview of deep learning (CNN, RNN, Autoencoder, graph neural network, etc.) 2. Overview of deep learning in Bioinformatics	Lecture, discussion
Week 4	1. Natural language processing (NLP, Transformer) 2. Gene ontology and knowledge graph 3. Explainable Artificial Intelligence (XAI)	Lecture, discussion Homework 1
Week 5	Cutting-edge AI topics: 1. Pre-trained large language models (BERT, GPT) 2. Generative AI (GAN, VAE, Diffusion models)	Lecture, discussion Projects start
Week 6	1. Overview of biological databases 2. Data visualization 3. Workflow 4. Cloud computing	Lecture, discussion



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Learning plan (continued)



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Schedule	Teaching Content	Teaching Form
Week 7	1. Introduction to sequence alignment 2. Pairwise alignment 3. Multiple sequence alignment (MSA)	Lecture, discussion Homework 2
Week 8	1. Next-generation sequencing (NGS) data 2. Genome-wide association study (GWAS) 3. Single-cell RNA-seq data analysis 4. Spatial transcriptomics	Lecture, discussion
Week 9	1. Proteomics 2. Epigenomics, metabolomics 3. Integration of multi-omics data	Lecture, discussion
Week 10	1. Overview of structural bioinformatics 2. Protein Data Bank (PDB) 3. AI for protein structure prediction and design (AlphaFold2, RFdiffusion)	Lecture, discussion Homework 3
Week 11	1. Overview of systems biology 2. Overview of synthetic biology	Lecture, discussion
Week 12	1. The pipeline of drug discovery and development 2. Overview of bioinformatics in drug discovery 3. AI for drug discovery and design	Lecture, discussion
Weeks 13-16	Course practice project	Group meetings and final presentations



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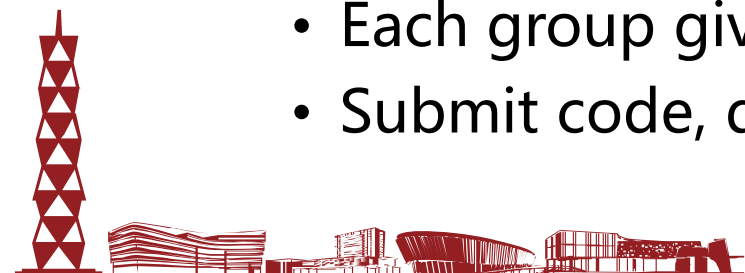
Prerequisites



- Required:
 - Some experiences in computer programming (mainly Python)
 - Knowledge of probability and statistics
- You' d better have:
 - Interests in coding
 - Interests in data science (esp. machine learning)
 - Interests in biology and medicine



- Class participation: **20%** (sign-in attendance sheet, quizzes)
- Homework: **30%**
 - 3 homework assignments
 - Individual based
 - Submit solutions to BlackBoard (BB) system of ShanghaiTech (specific instructions given later)
- Course project: **50%**
 - Group based (2-3 students)
 - To complete a data science project (using machine learning), or an algorithm implementation and analysis project
 - Mid-term evaluation in form of personal interview
 - Each group gives a final presentation of ~15 minutes
 - Submit code, data, PPT, etc. and a final report (**all in English**)





Policies for absence and late submission



- Attendance could be checked by quizzes (at random dates)
- If unable to attend a class for special reasons, you must:
 - Apply for approval of leave officially (in Egate and by email) **before** the class begins
 - Provide an official medical certificate (MC) from a clinician or other official document to justify your absence
- Penalty for late submission of homework or project solutions:
 - **50% deduction** within the first 24 hours after deadline
 - **100% deduction (0 score)** more than 24 hours after the deadline



Academic integrity rules



- Unless explicitly noted, your work submitted should be done **all by yourself** and reflect your **independent** capabilities.
- **No cheating:**
 - **No plagiarism** (i.e. copy any part of solution from others): We will scan your solutions for similarity both automatically and manually.
 - Do not ask anyone else (whether inside or outside the university) to do the homework or project on your behalf.
 - Do not show, explain (or share) your homework or project solutions to (with) any other student under any circumstance. In case of plagiarism, a student being copied shall be punished as severely as a student copying others.
 - If some sentences in your answer or report are from books, papers or web (e.g. Wikipedia), **cite the sources as references and add double quotations**. Otherwise, it is counted as plagiarism.
 - If a piece of source code in your project solution is borrowed or adapted from the web (e.g. GitHub), you should state clearly as comments in the code and/or in document how you have borrowed the code.
 - Do not violate academic integrity rules for quizzes and exam (further specific instructions will be given in due time).



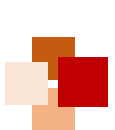


Academic integrity rules (continued)



- **Consequences of cheating**: 0 score for course component; fail the course; record in credit system; expulsion from the university.
- **Reproducibility**:
 - Solutions you submit should be **reproducible** independently by yourself as well as others.
 - Your program is expected to run again to output results consistent with you report.
 - An answer not reproducible is considered a **"fake solution"**, and will get 0 score.
- Use of AI for coding and writing (e.g. ChatGPT):
 - If a piece of code is written by AI, state it clearly by comments and in documents
 - You should always write the first draft **by yourself**, not by AI. However, AI can be used to check grammatical errors, improve quality of English writing, etc.





Operating system and programming languages



- You are recommended, but not required, to use **Linux**, because:
 - Most open-source software tools in bioinformatics are based on Linux (or other Unix-like systems)
 - Command-line and shell are powerful and flexible for sophisticated handling of big data
- Programming Languages (in the order of preference):
 - Python
 - R
 - Others: C/C++, Java, SQL, MATLAB, Perl, JavaScript, PHP, etc.



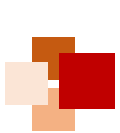


Course materials



- Main course site on BlackBoard (BB) of ShanghaiTech (教学互动平台), including:
 - Lecture PPT slides (in PDF; usually posted one day in advance; **minor revisions after posting is possible**)
 - Homework questions, solutions and scores
 - Project description
 - Etc.
- Note: Please do not share the course materials outside the class, to avoid potential **copyright issue**





Text books



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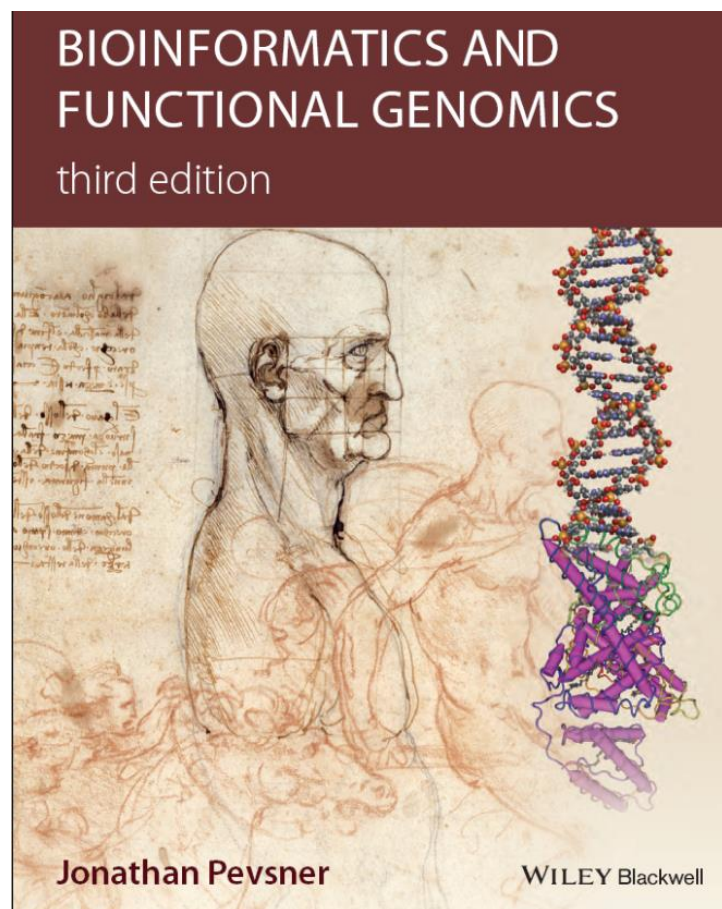
Book title: Bioinformatics and Functional Genomics (3rd Edition)

Author(s): Jonathan Pevsner

Publisher: John Wiley & Sons

Publication Year: 2015

ISBN: 978-1118581780



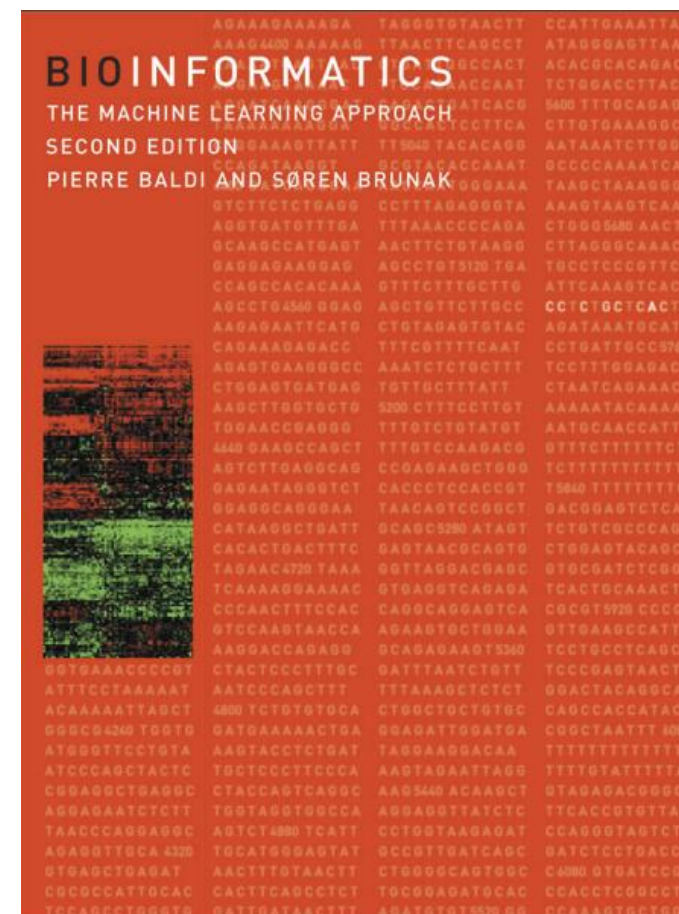
Book title: Bioinformatics: The Machine Learning Approach (2nd Edition)

Author(s): Pierre Baldi, Søren Brunak

Publisher: MIT Press

Publication Year: 2001

ISBN: 0-262-02506-X





Other references

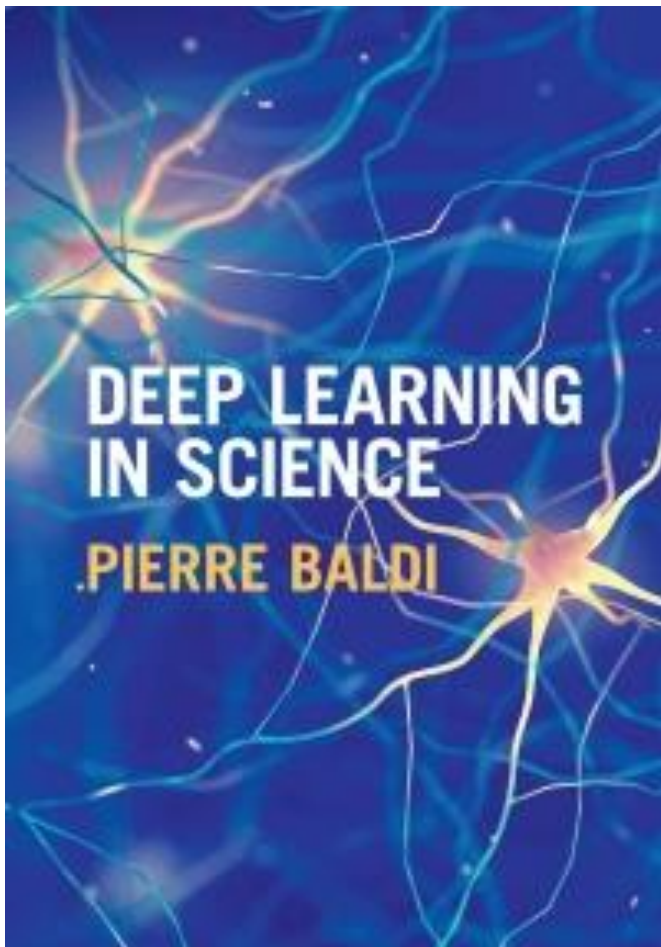
Book title: Deep Learning in Science

Author(s): Pierre Baldi

Publisher: Cambridge University Press

Publication Year: 2021

ISBN: 978-1-108-84535-9



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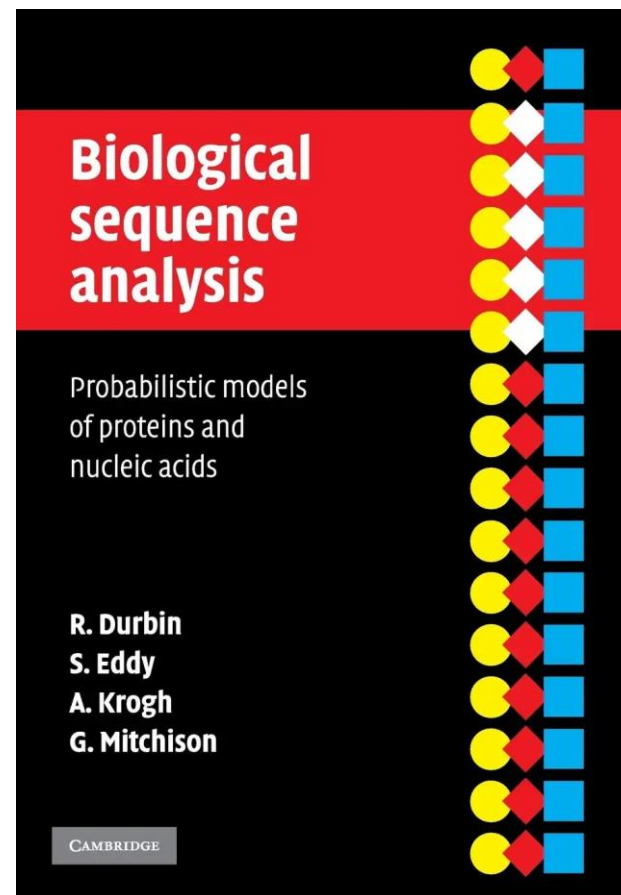
Book title: Biological Sequence Analysis: Probabilistic Models of Proteins and Nucleic Acids

Author(s): Richard Durbin, Sean R. Eddy, Andres Krogh, Graeme Mitchison

Publisher: Cambridge University Press

Publication Year: 1998

ISBN: 978-0521629713 (or 0521629713)





Communications



- Each student is required to check your ShanghaiTech email (as registered with the course) **at least once every 24 hours**. If missing any important email about the course, the student should bear the consequence by himself/herself
- A WeChat group will be set up, and you are expected to read the messages there in time
- For technical discussions:
 - After lectures, office hours or by appointment
 - Email and Wechat are only for short messages (e.g. making an appointment), but **not** for technical discussions
 - Use the BlackBoard system for Q&A
- The lecturer or TA shall **not** help any student debug his/her code
- Human languages:
 - Written: mostly in English, occasionally in Chinese
 - Spoken: mostly in Chinese, occasionally in English

